TreasuryAudit360 – Automated Audit of Bank Treasury (India)

Capstone project: n8n workflow automation + Claude AI + hosted web app

Workflow automation for audit of the Treasury department of a bank in India

Regulatory coverage: RBI | SEBI | IFSCA (GIFT City) | FEMA

Auditing framework: ICAI Standards on Auditing and Standards on Internal Audit

Platform: n8n Cloud | Claude (Anthropic) | Google Sheets | Gmail | Lovable

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Demo data notice: "Bharat Demo Bank Ltd" and every counterparty, dealer, ISIN and amount in this project are fictitious. Regulatory thresholds are stored as *parameters* and must be verified against the latest RBI / SEBI / IFSCA circulars before any real use.

1. What the solution does (end to end)

Stage	What happens	Audit standard / regulation
1. Data ingestion	Treasury extracts (deal blotter, investment register, counterparty & dealer limits, daily risk positions, SEBI restricted list, parameters) are read from Google Sheets	SA 500 – Audit evidence; SA 230 – data fingerprint logged
2. Risk-based testing	23 automated tests run on **100% of the population** (CAAT)	SA 315 (Revised), SA 330, SA 520, RBI RBIA framework
3. Regulatory mapping	Every exception is tagged to RBI / SEBI / IFSCA / FEMA and to an ICAI SA	SA 250 – Laws & regulations
4. Risk rating	High / Medium per finding; overall rating for the engagement; materiality from parameters	SA 320, SA 450
5. AI report drafting	Claude drafts the Audit Committee report (executive summary, compliance matrix, observations, SA 265 deficiencies, follow-up, limitations) using only the computed facts	SIA 370 (reporting), SA 265
6. Documentation	Findings register + audit-trail record (run id, preparer, reviewer, fingerprint) saved to Google Sheets	SA 230 – Audit documentation
7. Communication	Report emailed to the auditor (Gmail) and shown in the web app	SA 265, SIA 370
8. Follow-up loop	Management enters response, owner, target date, status for each finding in the web app → written back to the register	SIA 370 follow-up, SA 580 (written representations)
9. Continuous audit	Optional daily 7 pm schedule (concurrent audit)	RBI Concurrent Audit System guidelines

Architecture

```
Treasury system / CBS extracts
          │  (CSV / Excel → Google Sheets)
          ▼
Google Sheet "Treasury Audit - Demo Bank (Capstone)"
  Parameters · Deals · Investments · Counterparties · Dealers · Positions · Restricted_List
          │
          ▼
n8n Cloud ── Workflow 1: Audit Engine  (POST /webhook/treasury-audit/run, + daily schedule)
          │     Read 7 tabs → Run 23 tests (Code node) → Claude drafts report
          │     → Save Findings → Log Audit_Runs (SA 230) → Gmail report → JSON response
          │
          ├── Workflow 2: Findings & Data API
          │     GET  /webhook/treasury-audit/findings
          │     GET  /webhook/treasury-audit/data
          │     POST /webhook/treasury-audit/response   (management response)
          │
          ▼
Lovable web app "TreasuryAudit360" (hosted on lovable.app)
  Dashboard · Run Audit · Audit Report · Findings Register · Compliance Matrix · Data Explorer · Architecture
```

2. What has already been built for you

Item	Location
Workflow 0 – Setup Demo Data	https://nonchallant.app.n8n.cloud/workflow/lIUnOcDT7XaVmba2 (already run once – do **not** run again or a second workbook is created)
Workflow 1 – Audit Engine	https://nonchallant.app.n8n.cloud/workflow/UAWiMJKX6g0aDfu6

Item	Location
(published)	
Workflow 2 – Findings & Data API (published)	https://nonchallant.app.n8n.cloud/workflow/T1Z2qgsCgpxW9gob
Google Sheet (data + findings + audit trail)	https://docs.google.com/spreadsheets/d/1HogxieIBSdBVLSRC7ktPiUNNHaB2ruNFZ6teT0Rkuek/edit
Web app (Lovable editor)	https://lovable.dev/projects/81783906-7834-4b2a-b9f3-a475c9424449
Web app (live, public)	https://treasuryaudit360.lovable.app

Tested on 30-Oct-2026: the live webhook returned 31 findings (21 High, 10 Medium) across all 23 tests, a full Claude report, and the management-response update wrote back to the sheet.

3. Step-by-step: how to build it yourself in n8n (for your viva / report)

Step 3.1 – Accounts and credentials

1. Create an **n8n Cloud** account (n8n.io → Start free). Note your instance URL, e.g. `https://<name>.app.n8n.cloud`.
2. In n8n → **Credentials → Add credential**:
   - *Google Sheets OAuth2* – sign in with your Google account.
   - *Gmail OAuth2* – same Google account (used to email the report).
   - *Anthropic* – either your Anthropic API key, or n8n's built-in AI credits (shown as "Gateway credits" on the Anthropic node).
3. Create a **Lovable** account (lovable.dev) for the front end.

Step 3.2 – Prepare the data workbook (Workflow 0)

1. New workflow → add **Manual Trigger**.
2. Add **Google Sheets → Create spreadsheet** with sheets: `Parameters, Deals, Investments, Counterparties, Dealers, Positions, Restricted_List, Findings, Audit_Runs`.
3. Add one **Code** node per data tab that returns the sample rows (see the Seed nodes in Workflow 0), each followed by **Google Sheets → Append row** (Mapping: *Map automatically*; Cell format: *RAW* so dates/times stay as text).
4. Run once. Copy the spreadsheet ID from the output.

For a live audit, replace this step by exporting the treasury system reports (Murex / Calypso / Finacle Treasury / CBS) to the same column layout.

Step 3.3 – Parameters tab (makes the tool regulation-aware without code changes)

key	demo value	why
capital_funds_cr / net_worth_cr	12000 / 9500	base for call-money and capital-market limits
call_borrow_limit_pct / call_lend_limit_pct	125 / 50	RBI Call, Notice & Term Money Directions 2021 (per-day ceilings)
cme_limit_pct	40	RBI Exposure Norms – capital market exposure
noop_limit_usd_mn	150	Board-approved NOOP (RBI MD Risk Mgmt & Inter-Bank Dealings)
rate_tolerance_bps / price_tolerance_pct	10 / 0.25	off-market rate scan
confirmation_sla_hours	2	back-office SOP
dealing_start / dealing_end	09:00 / 17:30	dealing-room policy
htm_opening_book_cr / htm_sale_threshold_pct	18000 / 5	RBI Investment Directions 2023 – HTM sales disclosure
materiality_cr	5	SA 320 performance materiality
default_auditor_email	(blank)	used by the scheduled run

Step 3.4 – Build the Audit Engine (Workflow 1)

1. **Webhook** node – Method POST, Path `treasury-audit/run`, Respond: *Using 'Respond to Webhook' node*, Allowed origins *.
2. **Schedule Trigger** – every day 19:00 (keep disabled until needed).
3. **Edit Fields (Set) "Normalize Request"** – `auditor_name`, `auditor_email`, `engagement`, `trigger_source` from `$json.body` with defaults.
4. Seven **Google Sheets → Get row(s)** nodes in a chain (Parameters → Deals → Investments → Counterparties → Dealers → Positions → Restricted_List). In each node's *Settings* turn on **Execute Once** (so each tab is read only once).
5. **Code node "Run Audit Tests"** – holds the test catalogue (id, regulator, regulation, ICAI SA, default risk, recommendation) and the logic for all 23 tests (section 4). Outputs one item: run id, stats, tests, findings, compliance matrix, SA 530 walkthrough sample, data fingerprint and a compact JSON brief for the AI.
6. **Anthropic → Message a model** "Draft Audit Report with Claude" – model `claude-sonnet-5-5`, Max tokens 14000, a System prompt describing the CA-auditor role and the 8 required report sections, user message = the JSON brief. *Settings → On error: Continue* so a failed AI call never stops the audit.
7. **Code "Assemble Report"** – cleans the AI HTML, appends Annexure A (findings register) and Annexure B (tests performed) generated from data (so the report is always factually complete), builds the email subject and API response.
8. **Code "Prepare Findings Rows"** → **Google Sheets Append** to `Findings`.
9. **Set "Build Audit Trail Record"** (Execute Once) → **Google Sheets Append** to `Audit_Runs` (run id, time, preparer, *reviewed_by = Pending review*, rating, counts, data fingerprint, emailed to).
10. **Gmail → Send** (HTML) to the auditor email (Execute Once, On error: Continue).
11. **Respond to Webhook** – Respond with JSON `{{ $('Assemble Report').first().json.response }}`.
12. Click **Publish** (top right). The production URL is `https://<instance>/webhook/treasury-audit/run`.

Step 3.5 – Build the Findings & Data API (Workflow 2)

- `GET treasury-audit/findings` → Read `Findings` → Read `Audit_Runs` (Execute Once) → Code (filter latest run, summaries) → Respond JSON.
- `GET treasury-audit/data` → read all 7 data tabs (Execute Once each) → Code (one object with arrays) → Respond JSON.
- `POST treasury-audit/response` → Set (finding_id, status, response, owner, target date, timestamp) → **Google Sheets → Update row** matching column `finding_id` → Respond JSON.
- Publish.

Step 3.6 – Test with any HTTP client

```
curl -X POST https://nonchallant.app.n8n.cloud/webhook/treasury-audit/run -H "Content-Type:
application/json" -d "{\"auditor_name\":\"CA
Ravi\",\"auditor_email\":\"\",\"engagement\":\"Concurrent Audit - Treasury\"}"
```

Expect ~60–120 seconds (most of it is Claude writing the report).

4. Audit test catalogue (what the engine checks)

Test	Check	Regulator	Regulation (as configured)	ICAI Standard
TR-01	Unconfirmed / unmatched deals	RBI	MD – Risk Management & Inter-Bank	SA 505, SA

Test	Check	Regulator	Regulation (as configured)	ICAI Standard
			Dealings 2016; FEDAI; CCIL matching	330
TR-02	Confirmation beyond SLA	RBI	MD – Risk Mgmt & Inter-Bank Dealings; back-office SOP	SA 330, SA 315
TR-03	Dealer confirming own deal (front/back office segregation)	RBI	Internal-control guidelines for treasury	SA 315, SA 265, SA 240
TR-04	Off-market rate deals (yield bps / price %)	RBI / FEMA	MD – Risk Mgmt & Inter-Bank Dealings; FEDAI; FBIL/FIMMDA	SA 240, SA 520
TR-05	After-hours deals without approval / unrecorded lines	RBI	Dealing-room policy	SA 315, SA 330
TR-06	Counterparty exposure limit breach	RBI	Exposure Norms; Large Exposures Framework	SA 250, SA 330
TR-07	Dealer per-deal limit breach	RBI	Board treasury policy	SA 315, SA 265
TR-08	Stop-loss breach	RBI	Market risk policy	SA 315, SA 540
TR-09	NOOP breach	RBI / FEMA	MD – Risk Mgmt & Inter-Bank Dealings; FEMA 1999	SA 250, SA 330
TR-10	Call borrowing / lending above capital-fund ceilings	RBI	Call, Notice & Term Money Directions 2021	SA 250, SA 330
TR-11	Settlement failure / SGL bounce	RBI	SGL bouncing instructions; CCIL	SA 330, SA 505
TR-12	HTM sale without Board approval; HTM sales > 5% (disclosure)	RBI	Investment Portfolio Directions 2023	SA 250, SA 450
TR-13	Fair-value loss not recognised (AFS / FVTPL)	RBI	Investment Portfolio Directions 2023	SA 540, SA 450
TR-14	Stale valuation	RBI	Investment Portfolio Directions 2023	SA 540, SA 500
TR-15	Non-SLR below investment grade / unrated / unlisted	RBI	Investment Portfolio Directions 2023	SA 250, SA 540
TR-16	Capital market exposure ceiling	RBI	Exposure Norms	SA 250, SA 330
TR-17	Trade in restricted-list scrip (UPSI)	SEBI	SEBI (PIT) Regulations 2015	SA 250, SA 240
TR-18	User derivative without suitability / ISDA	RBI	Market-makers in OTC Derivatives Directions 2021	SA 250, SA 315
TR-19	IBU transaction in INR	IFSCA	IFSCA (Banking) Regulations 2020; FEMA	SA 250, SA 330
TR-20	IBU ring-fencing (desk vs book mismatch)	IFSCA	IFSCA (Banking) Regulations 2020	SA 315, SA 265
TR-21	IBU deal with resident individual	IFSCA / FEMA	IFSCA Banking Regulations; FEMA (LRS)	SA 250
TR-22	Deals with KYC-pending counterparty	RBI	KYC Master Direction 2016; PMLA	SA 250, SA 240
TR-23	Holdings vs NSDL/CDSL/SGL reconciliation	SEBI / RBI	SEBI (Depositories & Participants) Regs 2018; SGL/CSGL	SA 505, SA 500

Other ICAI standards applied across the engagement: SA 200 (overall objectives), SA 230 (documentation – audit trail tab and data fingerprint), SA 265 (deficiency reporting – report section), SA 320 (materiality parameter), SA 530 (top-5 walkthrough sample for vouching), SA 580 (management representations – response loop), SIA 370 (internal audit reporting).

5. Hosting – options and steps

Chosen design (recommended for the capstone):

- **Workflows:** n8n Cloud (managed hosting; HTTPS webhooks included). Nothing else to deploy – publishing a workflow makes its webhooks live.
- **Front end:** Lovable (builds and hosts the React app on `*.lovable.app`).
- **Data store:** Google Sheets (easy for examiners to inspect).

The web app is already published at https://treasuryaudit360.lovable.app. After any change in the Lovable editor, click top-right **Publish → Update** to push it live.

Alternative – self-host n8n (if your evaluator wants on-premise / data-localisation):

1. Get a VPS (AWS Mumbai `ap-south-1`, Azure Central India, or DigitalOcean Bangalore) – data stays in India, relevant for RBI data-localisation expectations.
2. Install Docker, then:

```
   docker volume create n8n_data
   docker run -d --name n8n -p 5678:5678 -e N8N_HOST=audit.yourdomain.in -e
WEBHOOK_URL=https://audit.yourdomain.in/ -e GENERIC_TIMEZONE=Asia/Kolkata -v n8n_data:/home/node/.n8n
docker.n8n.io/n8nio/n8n
```

3. Put Caddy or Nginx in front for HTTPS; restrict the webhooks with Header Auth and an IP allow-list.
4. Import the three workflow JSON files (n8n → Workflows → ⋯ → Import from file), re-select credentials, publish.
5. Change `BASE` in the Lovable app's `src/config.ts` to your new URL.

Export your workflows for submission: open each workflow → ⋯ (top right) → **Download** → saves a `.json` file. Attach the three JSON files to your project.

6. Demo video script (~8–10 minutes)

Time	Screen	What to say / do
0:00	Title slide or app home	"TreasuryAudit360 automates concurrent / internal audit of a bank's treasury, covering RBI, SEBI, IFSCA (GIFT City) and FEMA, and documented per ICAI Standards on Auditing."
0:45	**About / Architecture** page	Walk the flow: data → n8n → 23 tests → Claude → register → email → management response.
1:30	**Google Sheet**	Show tabs Deals, Investments, Positions, Restricted_List and the **Parameters** tab: "Regulatory limits are parameters – when RBI changes a limit, the auditor edits one cell."
2:15	**n8n – Audit Engine** canvas	Explain the 3 groups: Load data → Audit & report → Record, notify & respond. Open the "Run Audit Tests" Code node briefly to show the test catalogue with regulation and SA references.
3:15	**Web app → Run Audit**	Enter your name and email, click Run. While stages animate: "100% population testing – SA 530 sampling is replaced by CAAT; a top-5 walkthrough sample is still selected for vouching."
4:30	Results	Overall rating, 31 findings, compliance matrix, tests table, SA 530 sample.
5:15	**Audit Report** page	Scroll the Claude-drafted Audit Committee report: executive summary, compliance matrix, observations by regulator, SA 265 deficiencies, follow-up, limitations; then Annexures A & B. Click Print / Save as PDF.
6:15	**Gmail inbox**	Show the emailed report.
6:45	**Findings Register**	Filter Regulator = IFSCA (GIFT City ring-fencing & INR deal), SEBI (insider-list trade). Open TD006 (dealer confirmed own deal) → enter management response, owner, target date, status "Action Initiated" → Submit.
7:45	Google Sheet → Findings tab	Show the response written back; Audit_Runs tab = SA 230 audit trail (preparer, reviewer pending, data fingerprint).
8:15	**Compliance Matrix** + **Data Explorer**	Regulator-wise status; population tested.
8:45	n8n Executions list + Schedule node	"Every run is logged; enable the daily schedule for continuous concurrent audit."
9:15	Close	Benefits (100% coverage, speed, consistency, documentation), limitations (data quality, AI narrative reviewed by auditor, parameters to be verified), future scope.

Recording tips: run one audit *before* recording so the dashboard is populated; record the live run anyway (it takes ~1–2 min – cut the wait in editing). Use OBS Studio or Windows `Win + Alt + R` (Xbox Game Bar), 1080p.

7. Controls, limitations and disclaimers (put in your report)

- The AI only narrates facts computed by deterministic tests; the auditor must review the narrative before issuing (professional judgement under SA 200 / SA 220).
- Results depend on completeness and accuracy of data supplied by management (SA 500, SA 580).
- Thresholds are configurable and illustrative; verify against current RBI Master Directions/Circulars, SEBI regulations and IFSCA regulations.
- Webhooks in the demo are open (no authentication). For production: add Header Auth/JWT, IP allow-listing, self-hosting in India, encrypted credentials, and role-based access in the web app.
- Not covered in the demo (future scope): CRR/SLR maintenance from Form A/VIII returns, LCR/NSFR, AGL and PV01 limits, FEMA/RBI regulatory return reconciliation (e.g. XBRL/CIMS), voice-log sampling, Ind AS 109 readiness.

8. Re-running the demo cleanly

- Each run appends a new set of findings and an Audit_Runs row – the app always shows the latest run.
- To reset, delete rows 2+ in the `Findings` and `Audit_Runs` tabs (keep the header row).
- To show a "clean" audit, fix data in the sheet (e.g. set TD006 `confirmed_by` = B201) and run again – that exception disappears.